

THE DEPARTMENT OF PHYSICS KHS

Names:

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ITEMS	LEARNERS SCORES
Item 1	
Item 2	
Total Scores	

KIYALA HIGH SCHOOL

END OF YEAR ASSESSMENT TEST 2025

PHYSICS

S.3

Time: 1 Hours

INSTRUCTIONS:

- The paper contains **two** items.
- Attempt **all the two** items from the paper.

ITEM 1:

A certain carpenter who was hired to roof the recreation hall near a tall tree which was next to the pool of water. The hammer of the carpenter accidentally fell into the pool of water. The carpenter tried to clap the hands once to alert the helper who was down to pick the hammer before reaching the pool of water. To his surprise, two claps were heard instead of one, and to make it worse as the carpenter continued clapping, the clapping became confused, unclear and indistinct.

On looking down the pool, the carpenter realised that the hammer was appearing near the surface, when tried to pick it, it was not possible. At around mid-day after work, as the carpenter was going back home, a pool of water was seen appearing at the middle of the road. This even confused the carpenter the more.

Hint

- **Speed of sound in air = 340ms^{-1}**
- **The carpenter heard the second clap after 0.5 seconds**

Task:

As a student of Physics, help the carpenter understand:

- Why two claps were heard by the carpenter.
- The clapping became confused and unclear and how this can be improved.

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- c) The carpenter was not able to reach where the hammer was despite it being nearer the surface.
- d) The carpenter noticed a pool of water on the road.
- e) To calculate how far the carpenter from the tall tree.

ITEM 2:

In a certain area, a farmer decided to be fetching water from downstream of the water body for watering the crops during the dry season. The farmer had two types of shoes, spiked and flat shoes. On a particular day when the farmer used spiked shoes, the farmer found it difficult to walk and started sweating and got tired compared to when the flat shoes were used.

The farmer was later advised to acquire a tank of capacity 10,000 litres to help in watering the crops. After filling the tank with water to a height of 1.80 m , the farmer was confused whether to put the tap either at the top or at the bottom of the tank. Temperature of the water in the tank raised from 35°C to 50°C since the tank was left under the sun in the open.

Hint

- ***Acceleration due to gravity = 10ms^{-2}***
- ***Density of water = 1000kgm^{-3}***
- ***Specific heat capacity of water = $4200\text{Jkg}^{-1}\text{K}^{-1}$***
- ***Mass of water in the tank = $10,000\text{kg}$***

Task:

As a student of Physics, help the farmer to understand:

- a) As to why he got tired when using spiked shoes and why he got sweat.
- b) To determine the pressure of water in the tank.
- c) Design the tank by putting the tap in the right position.
- d) Calculate the heat generated in the tank.

**WISHING YOU MERY CHRISMAS AND PROSPEROUS NEW YEAR
2026.**

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